Cardiology II: Heart Failure, Arrhythmias and Atrial Fibrillation

All key points for the exam (AHA/ACC/HFSA 2022 HF + 2024 ECDP, ESC 2023, ACC/AHA/HRS 2023 AF) — drug tables included

PART A — HEART FAILURE (HF)

1. Definitions, Classification and Diagnosis

- **HF** = clinical syndrome from structural/functional impairment of ventricular filling or ejection → **congestion** and/or **low cardiac output** • **neurohormonal activation** (**RAAS**, **sympathetic nervous system**, vasopressin) is initially compensatory but causes **remodeling** and progression → the targets of GDMT.
- **Causes: CAD/MI** (most common in HFrEF), **hypertension** (most common in HFpEF), valvular disease, cardiomyopathies (dilated, hypertrophic, alcohol, peripartum), myocarditis, arrhythmias (tachycardia-induced), thyroid disease, **drugs/toxins** (anthracyclines, trastuzumab, alcohol, cocaine).
- **Precipitants of decompensation: non-adherence** (medications, **sodium/fluid**), uncontrolled HTN, **ischemia/ACS**, **AF/arrhythmias**, infection, renal failure, anemia, thyroid disease, **drugs** (NSAIDs, non-DHP CCBs, TZDs).

Classification	Definition
HFrEF	**LVEF ≤ 40%** (systolic dysfunction)
HFmrEF (mildly reduced)	**LVEF 41–49%**
HFpEF	**LVEF ≥ 50%** (diastolic dysfunction) + evidence of ↑ filling pressures • older, women, HTN, obesity, diabetes, AF
HFimpEF (improved)	Previous LVEF ≤ 40% and follow-up LVEF > 40% (≥ 10-point increase) → **continue GDMT** (stopping → relapse)

ACC/AHA stage	Description	NYHA functional class
A — at risk	Risk factors (HTN, DM, CAD, obesity, cardiotoxins, family history) — no structural disease or symptoms	—
B — pre-HF	Structural heart disease/↑ filling pressures/↑ biomarkers **without symptoms** (e.g., prior MI, LVH, ↓ LVEF)	I (no limitation)
C — symptomatic HF	Structural disease **with current or prior symptoms**	I–IV
D — advanced HF	Marked symptoms interfering with daily life despite GDMT, recurrent hospitalizations	IV

- **NYHA: I** — no limitation with ordinary activity • **II** — slight limitation (symptoms with ordinary activity) • **III** — marked limitation (symptoms with less than ordinary activity) • **IV** — symptoms at rest • **stages never go backward; NYHA class can improve or worsen**.
- **Symptoms/signs: congestion** — dyspnea on exertion, **orthopnea**, **paroxysmal nocturnal dyspnea (PND)**, pulmonary crackles/edema, **peripheral edema**, **JVD**, hepatojugular reflux, **weight gain**, ascites, S3 gallop • **low output** — fatigue, weakness, cold extremities, confusion, ↓ urine output, hypotension.
- **Diagnosis: echocardiography** (LVEF, structure) • **natriuretic peptides**: **BNP ≥ 35–100 pg/mL** or **NT-proBNP ≥ 125–300 pg/mL** support HF (↑ in renal failure, age, AF; ↓ in obesity) • **sacubitril/valsartan ↑ BNP** (neprilysin substrate) → **use NT-proBNP** for monitoring • ECG, CXR, BMP, CBC, TSH, iron studies.

2. Treatment of HFrEF (Stage C) — Guideline-Directed Medical Therapy (GDMT)

Four pillars that ↓ mortality and hospitalization — start all as early as possible (within weeks, even simultaneously at low doses), titrate to **target doses** every 1–2 weeks: **(1) ARNI (or ACEi/ARB) • (2) evidence-based β-blocker • (3) MRA • (4) SGLT2 inhibitor**. **Loop diuretics** for congestion (symptoms only, no mortality benefit).

Drug	Starting dose	Target dose
Sacubitril/valsartan (ARNI)	**49/51 mg BID** (24/26 mg BID if ACEi/ARB-naive or low dose, eGFR < 30, moderate hepatic impairment, elderly)	**97/103 mg BID**
Lisinopril	2.5–5 mg daily	20–40 mg daily
Enalapril	2.5 mg BID	10–20 mg BID
Captopril	6.25 mg TID	50 mg TID
Ramipril	1.25–2.5 mg daily	10 mg daily
Losartan	25–50 mg daily	50–150 mg daily
Valsartan	20–40 mg BID	160 mg BID
Candesartan	4–8 mg daily	32 mg daily
Carvedilol	3.125 mg BID	25 mg BID (**50 mg BID if > 85 kg**)
Carvedilol CR	10 mg daily	80 mg daily
Metoprolol succinate (ER)	12.5–25 mg daily	**200 mg daily**
Bisoprolol	1.25 mg daily	10 mg daily
Spironolactone	12.5–25 mg daily	25–50 mg daily
Eplerenone	25 mg daily	50 mg daily
Dapagliflozin	10 mg daily	10 mg daily
Empagliflozin	10 mg daily	10 mg daily
Hydralazine + isosorbide dinitrate	Fixed (BiDil) 20/37.5 mg: 1 tab TID • or H 25–50 mg TID + ISDN 20–30 mg TID	2 tabs TID (40/75 mg) • H 300 mg/day + ISDN 120 mg/day
Ivabradine	2.5–5 mg BID	7.5 mg BID (titrate to HR 50–60)
Vericiguat	2.5 mg daily	10 mg daily
Digoxin	0.125–0.25 mg daily	Serum level **0.5–0.9 ng/mL**

(1) RAAS inhibition — ARNI, ACEi, ARB

- **ARNI (sacubitril/valsartan, Entresto) is preferred** (PARADIGM-HF: ↓ mortality vs enalapril) — as **first-line** or switch from ACEi/ARB in NYHA II–III • ACEi if ARNI not possible; **ARB** if ACEi intolerant (cough) or ARNI not possible.
- **Sacubitril** inhibits **neprilysin** → ↑ natriuretic peptides, bradykinin (vasodilation, natriuresis) • combined with ARB (valsartan) — **never with an ACEi**.
- **Switching ACEi → ARNI: 36-hour washout** (angioedema risk) • ARB → ARNI: no washout (start at next dose).
- **Contraindications: history of angioedema** (ACEi/ARNI), **pregnancy** (boxed — fetal toxicity), **bilateral renal artery stenosis**, concomitant **aliskiren in diabetes**, ARNI with ACEi (36 h).
- **ADRs: hypotension** (most common with ARNI), **hyperkalemia**, **↑ SCr** (↑ up to 30% acceptable — don't stop), **cough** (ACEi — bradykinin; switch to ARB), **angioedema** (ACEi > ARNI > ARB; higher in Black patients) • check BMP 1–2 weeks after start/titration.

(2) β-blockers

- **Only 3 evidence-based β-blockers in HFrEF: carvedilol, metoprolol succinate (ER), bisoprolol** (↓ mortality ~35%) • **metoprolol tartrate** and atenolol are **not** recommended.
- **"Start low, go slow"** — double the dose every 2 weeks • start only when **euvolemic/stable** (not in acute decompensation or shock) • initial transient fatigue/fluid retention (↑ diuretic) • benefits take months.
- **Continue during hospitalization for decompensation** unless cardiogenic shock/hypotension (reduce dose if needed) • abrupt withdrawal → rebound.
- **Carvedilol** = nonselective β + **α1 blockade** (more BP lowering, take **with food** to ↓ orthostasis) • metoprolol succinate/bisoprolol = β1-selective (better in asthma/COPD).
- **ADRs:** bradycardia, hypotension, fatigue, dizziness, worsening HF initially, masked hypoglycemia, bronchospasm, erectile dysfunction.

(3) Mineralocorticoid receptor antagonists (MRA)

- **Spironolactone or eplerenone** for **NYHA II–IV** with LVEF ≤ 35–40% (RALES, EMPHASIS-HF) — if **eGFR > 30** and **K < 5.0 mEq/L**.
- Monitor **K and SCr** at baseline, **3 days, 1 week**, then monthly × 3 months, then every 3 months • **stop if K > 5.5** • avoid K supplements/salt substitutes, NSAIDs.
- **Spironolactone: gynecomastia**, breast tenderness, impotence, menstrual irregularities (antiandrogenic) → switch to **eplerenone** (selective) • **eplerenone:** CYP3A4 substrate (**CI with strong inhibitors** — ketoconazole, clarithromycin, ritonavir).
- **Hyperkalemia risk** ↑ with ACEi/ARB/ARNI, CKD, diabetes, elderly, TMP-SMX → potassium binders (patiromer, SZC) may enable GDMT.

(4) SGLT2 inhibitors

- **Dapagliflozin 10 mg** or **empagliflozin 10 mg** daily — **regardless of diabetes** (DAPA-HF, EMPEROR-Reduced) → ↓ CV death and HF hospitalization • start if **eGFR ≥ 20–25** • sotagliflozin (SGLT1/2) also approved.
- Also **recommended for HFmrEF and HFpEF** (EMPEROR-Preserved, DELIVER) • can be started in hospital.
- ADRs: **genital mycotic infections**, volume depletion (may ↓ loop diuretic dose), **euglycemic DKA** (hold 3 days before surgery/sick days), UTI, small initial eGFR dip.

Additional therapies (selected patients)

Therapy	Indication	Notes
Hydralazine + isosorbide dinitrate (H-ISDN)	**Self-identified Black patients** with NYHA III–IV on optimal GDMT (A-HeFT: ↓ mortality) • or patients who **cannot take ACEi/ARB/ARNI** (renal failure, hyperkalemia, angioedema)	Hydralazine (arteriolar dilator ↓ afterload) + nitrate (venodilator ↓ preload, NO donor) • TID dosing (adherence) • ADRs: headache, dizziness, hypotension, reflex tachycardia, **drug-induced lupus (hydralazine)** • CI with PDE-5 inhibitors
Ivabradine (Corlanor)	NYHA II–III, **LVEF ≤ 35%**, **sinus rhythm**, **resting HR ≥ 70 bpm** on **maximally tolerated β-blocker** (SHIFT)	Inhibits **I(f) current in SA node** → ↓ HR only • ADRs: **bradycardia**, **luminous phenomena (phosphenes)**, AF, fetal toxicity • CYP3A4 substrate (avoid strong inhibitors, grapefruit, diltiazem/verapamil) • ↓ HF hospitalization (no mortality)
Digoxin	Persistent symptoms despite GDMT (↓ **hospitalizations**, **no mortality benefit** — DIG trial) • AF rate control in HFrEF	See digoxin box below
Vericiguat (Verquvo)	**Worsening HF** (recent hospitalization/IV diuretic) with LVEF < 45% on GDMT (VICTORIA)	Soluble guanylate cyclase stimulator • hypotension, anemia • **CI with PDE-5 inhibitors, nitrates, riociguat** • **teratogenic** (REMS)
IV iron (ferric carboxymaltose, ferric derisomaltose)	**Iron deficiency** (ferritin < 100, or 100–299 with TSAT < 20%) with NYHA II–III	↑ functional status, ↓ hospitalizations • **oral iron not effective** • ESAs not recommended for anemia in HF
Omega-3 PUFA	NYHA II–IV (reasonable)	Modest benefit
ICD (primary prevention)	**LVEF ≤ 35%**, NYHA II–III, on GDMT ≥ 3 months, > 40 days post-MI, life expectancy > 1 year • LVEF ≤ 30% NYHA I (ischemic)	Prevents sudden cardiac death • wearable defibrillator as bridge
CRT (biventricular pacing)	**LVEF ≤ 35%**, sinus rhythm, **LBBB with QRS ≥ 150 ms**, NYHA II–IV on GDMT	↓ mortality and symptoms
Advanced (Stage D)	Refractory symptoms	Inotropes (palliative/bridge), LVAD, heart transplant, palliative care

Digoxin

- **MOA:** inhibits **Na^+/K^+-ATPase** → ↑ intracellular Ca → **positive inotrope** + ↑ vagal tone → **↓ AV node conduction** (rate control).
- **Dose: 0.125–0.25 mg daily** (0.125 mg or every other day in elderly, CKD, low weight) • **no loading dose in HF** • renally cleared (long half-life ~36–48 h; ↑ in CKD) • target level **0.5–0.9 ng/mL** in HF (higher levels ↑ mortality) • draw level ≥ 6–8 h after dose (steady state 1–3 weeks).
- **Toxicity** (narrow therapeutic index): **GI** (anorexia, **nausea, vomiting** — early sign), **CNS** (confusion, fatigue), **visual disturbances** (**yellow-green halos**, blurred vision), **arrhythmias** (any — PVCs, bigeminy, **bidirectional VT**, paroxysmal atrial tachycardia with block, AV block, bradycardia) • **hyperkalemia** in acute overdose.
- **↑ Toxicity risk: hypokalemia** (most important — diuretics), **hypomagnesemia**, **hypercalcemia**, hypothyroidism, renal impairment, elderly, ischemia.
- **DDIs ↑ digoxin levels: amiodarone** (↓ digoxin dose by **50%**), **dronedarone** (↓ 50%), **verapamil**, diltiazem, **quinidine**, propafenone, macrolides (clarithromycin), azoles, cyclosporine, ranolazine, ticagrelor (P-gp inhibitors) • ↓ levels: rifampin, St John's wort, cholestyramine, antacids.
- **Antidote: digoxin immune Fab (DigiFab)** — life-threatening arrhythmias, K > 5–5.5 in acute overdose, end-organ dysfunction • levels unreliable after Fab • correct K/Mg • avoid calcium in hyperkalemia from digoxin toxicity (historical caution).

Loop diuretics (congestion)

- **Furosemide** (oral bioavailability **~50%**, variable 10–100%), **bumetanide** (~80–100%), **torsemide** (~80–100%, long-acting) • **ethacrynic acid** = non-sulfonamide (**sulfa allergy**; more ototoxic).

- **Equivalent doses: furosemide 40 mg PO = furosemide 20 mg IV = torsemide 20 mg = bumetanide 1 mg**.
- Use the **lowest dose to maintain euvolemia** (dry weight) • **diuretic resistance** → ↑ dose, IV, switch to torsemide/bumetanide, or add **thiazide-type** (**metolazone 2.5–5 mg** 30 min before loop — sequential nephron blockade; watch hypokalemia/hyponatremia) or acetazolamide.
- **ADRs: hypokalemia**, **hypomagnesemia**, hyponatremia, **hypovolemia/prerenal AKI**, hypotension, **metabolic alkalosis**, hyperuricemia (gout), hyperglycemia, **ototoxicity** (rapid IV high doses, with aminoglycosides), photosensitivity, sulfa cross-sensitivity (low).
- **NSAIDs ↓ diuretic effect** and worsen HF/renal function.

Drugs to AVOID in HFrEF (cause fluid retention, negative inotropy, or arrhythmia)

- **NSAIDs** and COX-2 inhibitors (Na/water retention, ↓ diuretic effect, AKI) • **non-DHP CCBs** (**diltiazem, verapamil** — negative inotropes; amlodipine/felodipine OK) • **thiazolidinediones** (pioglitazone — fluid retention; CI NYHA III–IV) • **DPP-4i saxagliptin, alogliptin** • **class I antiarrhythmics** (flecainide, propafenone, disopyramide) • **dronedarone** (CI in NYHA IV or recent decompensation) • **itraconazole** (negative inotrope — boxed), **cilostazol** (CI in HF of any severity — PDE3), **anthracyclines**, trastuzumab, high-dose TNF-α inhibitors, **pregabalin**, minoxidil, **alcohol**, stimulants, sodium-containing products (effervescent tablets), **metformin is OK** (avoid if unstable/hypoperfusion).

3. HFmrEF, HFpEF and HFimpEF

- **HFpEF (LVEF ≥ 50%): SGLT2 inhibitor** (dapagliflozin/empagliflozin — recommended) • **diuretics** for congestion • **treat HTN** (BP < 130/80), AF, CAD, obesity, diabetes, OSA • **MRA** (spironolactone — may benefit; **finerenone** now approved for HFmrEF/HFpEF — FINEARTS-HF) • **ARNI or ARB** (may be considered, esp. LVEF at the lower end) • **GLP-1 RA/tirzepatide** for obese HFpEF (↓ symptoms, ↑ function) • exercise training • avoid routine nitrates/PDE-5 inhibitors.
- **HFmrEF (41–49%): SGLT2i** + diuretics • consider ARNI/ACEi/ARB, β-blocker, MRA (especially LVEF near 40%).
- **HFimpEF: continue all GDMT** (TRED-HF: withdrawal → relapse).

4. Acute Decompensated Heart Failure (ADHF)

Hemodynamic profile	Congestion?	Perfusion?	Treatment
Warm and dry (A)	No	Adequate	Optimize oral GDMT (compensated)
Warm and wet (B) — **most common**	**Yes** (↑ PCWP)	Adequate	**IV loop diuretics ± vasodilators** (nitroglycerin, nitroprusside) if BP adequate
Cold and wet (C)	Yes	**Low** (↓ CI)	**Inotropes** (dobutamine, milrinone) ± vasopressors if hypotensive + diuretics once perfusion restored • vasodilators if SBP allows
Cold and dry (L)	No	Low	Cautious fluids, then inotropes if needed

- **IV loop diuretic** dose = **1–2.5 × the total daily oral dose** given IV (bolus or continuous infusion — equivalent; DOSE trial) • monitor urine output (goal > 100–150 mL/h early), daily weights, BMP (K, Mg, SCr) • ↑ dose or add thiazide/acetazolamide if inadequate.
- **Vasodilators** (SBP > 90–100): **IV nitroglycerin** (venodilator → ↓ preload; tachyphylaxis 24–48 h, headache) • **nitroprusside** (balanced arterial + venous; **cyanide toxicity** — hepatic dysfunction, prolonged/high doses; **thiocyanate toxicity** in renal failure; protect from light; arterial line) • nesiritide (rarely used).
- **Inotropes** (cold, low output, end-organ hypoperfusion): **dobutamine** (β1 agonist; tachycardia, arrhythmias, ischemia; less effective with β-blockers) • **milrinone** (PDE3 inhibitor — inodilator; **hypotension**, arrhythmias, **renal dose adjustment**; works with β-blockers) • short-term only (↑ mortality) except palliative/bridge.
- **Continue GDMT** (β-blocker, ACEi/ARB/ARNI) unless **hypotension, cardiogenic shock, hyperkalemia, or AKI** • **start SGLT2i before discharge** • transition to oral diuretic 24 h before discharge • VTE prophylaxis • treat precipitants • morphine not recommended routinely • O2 if

hypoxemic, NIPPV for pulmonary edema.
- **Discharge:** GDMT optimized, follow-up within 7–14 days, education.

HF patient education

- **Daily weights** (same time, after voiding, before breakfast) → **report gain of > 2–3 lb (1 kg) in a day or 5 lb (2 kg) in a week**.
- **Sodium restriction** (< 2–3 g/day; avoid excessive) • fluid restriction (1.5–2 L/day) only in hyponatremia/advanced HF • limit alcohol • stop smoking.
- **Avoid NSAIDs**, decongestants, herbal products • adherence • vaccinations (influenza, pneumococcal, COVID) • exercise/cardiac rehab • recognize worsening symptoms.

5. Heart Failure Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Sacubitril/valsartan	**49/51 → 97/103 mg BID** (24/26 start if low-dose/naive, eGFR < 30)	**Hypotension**, hyperK, ↑ SCr, angioedema	**36-h washout from ACEi** • CI pregnancy, angioedema • ↑ BNP (use NT-proBNP)
ACE inhibitors	Lisinopril 2.5–5 → 20–40 mg • enalapril 2.5 → 10–20 BID	**Cough**, **angioedema**, hyperK, ↑ SCr	CI pregnancy, bilateral RAS • ↑ K with MRA, K supplements, TMP-SMX • NSAIDs ↓ effect
ARBs	Losartan 25–150 • valsartan 40–160 BID • candesartan 4–32	HyperK, ↑ SCr, hypotension	ACEi cough alternative • CI pregnancy
Carvedilol	**3.125 → 25 mg BID** (50 BID if > 85 kg)	Hypotension (α1), bradycardia, fatigue	With food • start when euvolemic
Metoprolol succinate	**12.5–25 → 200 mg daily**	Bradycardia, fatigue	β1-selective • not tartrate • CYP2D6
Bisoprolol	**1.25 → 10 mg daily**	Bradycardia	β1-selective
Spironolactone	**12.5–25 → 50 mg**	**Hyperkalemia**, **gynecomastia**	K < 5.0, eGFR > 30 • monitor K 3 d, 1 wk
Eplerenone	25 → 50 mg	HyperK	No gynecomastia • **CYP3A4** (CI strong inhibitors)
Dapagliflozin / empagliflozin	**10 mg daily**	Genital infections, volume depletion, euDKA	All LVEF • eGFR ≥ 20–25 • hold 3 days pre-op
Furosemide	20–40 mg PO daily–BID (max 600 mg/day)	**↓ K, ↓ Mg**, ↓ Na, AKI, ototoxicity, gout	40 PO = 20 IV • NSAIDs ↓ effect
Torsemide / bumetanide	10–20 mg / 0.5–1 mg daily (max 200 / 10)	Same	Better bioavailability
Metolazone	2.5–10 mg (30 min before loop)	**Severe ↓ K, ↓ Na**	Diuretic resistance
Hydralazine/ISDN	20/37.5 mg TID → 40/75 mg TID	Headache, dizziness, lupus (hydralazine)	Black patients NYHA III–IV • CI PDE-5i
Ivabradine	**2.5–5 → 7.5 mg BID**	**Bradycardia**, **phosphenes**, AF	Sinus rhythm, HR ≥ 70 on max β-blocker • CYP3A4 • teratogenic
Digoxin	**0.125–0.25 mg daily** (level 0.5–0.9)	**N/V, visual (yellow-green)**, arrhythmias	**Amiodarone/dronedarone → ↓ dose 50%** • ↑ with verapamil, quinidine • hypoK ↑ toxicity • DigiFab
Vericiguat	2.5 → 10 mg daily	Hypotension, anemia	Worsening HF • CI PDE-5i, nitrates, riociguat • REMS (teratogen)
Ferric carboxymaltose	750 mg IV × 2 (or weight/Hb-based)	**Hypophosphatemia**, HTN, hypersensitivity	Iron deficiency in HF
Dobutamine	2–20 mcg/kg/min	Tachycardia, arrhythmias	Cold/wet • less effect with β-blockers

Drug	Dose	Key ADRs	Key notes / interactions
Milrinone	0.125–0.75 mcg/kg/min	**Hypotension**, arrhythmias	Renal adjust • works with β-blockers
Nitroprusside	0.3–10 mcg/kg/min	Hypotension, **cyanide/thiocyanate toxicity**	Protect from light • renal/hepatic caution
Nitroglycerin IV	5–200 mcg/min	Headache, hypotension, tolerance	Warm/wet with adequate BP
Patiromer / SZC	8.4 g daily / 10 g TID → 5–10 g daily	HypoMg, constipation / edema	Enable RAAS inhibitor/MRA use

PART B — ARRHYTHMIAS

6. Antiarrhythmic Drug Classification (Vaughan Williams)

Class	Mechanism	Drugs	Key points
Ia	Block Na^+ (moderate) + K^+ → ↑ QRS, ↑ **QT**	**Quinidine**, **procainamide**, **disopyramide**	Quinidine: diarrhea, cinchonism, ↑ digoxin • procainamide: **drug-induced lupus**, active metabolite **NAPA** (renal), agranulocytosis • disopyramide: **anticholinergic**, negative inotrope (HCM use)
Ib	Block Na^+ (weak, fast dissociation) — ischemic tissue	**Lidocaine**, **mexiletine**	Ventricular arrhythmias only • lidocaine: CNS toxicity (confusion, seizures), hepatic metabolism • mexiletine: GI, tremor
Ic	Block Na^+ (strong) → ↑ QRS	**Flecainide**, **propafenone**	**CI in structural heart disease**, CAD/prior MI, HFrEF (**CAST — ↑ mortality**) • pill-in-pocket for AF • give with AV-nodal blocker (atrial flutter 1:1 conduction) • propafenone has β-blocking activity, metallic taste
II	**β-blockers** → ↓ SA/AV node	Metoprolol, esmolol, propranolol, atenolol, carvedilol	Rate control, SVT, post-MI, HF
III	**Block K^+** → ↑ action potential duration, ↑ **QT**	**Amiodarone**, **dronedarone**, **sotalol**, **dofetilide**, **ibutilide**	Torsades risk (except amiodarone low) • see details
IV	**Non-DHP CCBs** → ↓ AV node	Verapamil, diltiazem	Rate control, SVT • avoid HFrEF
Others	—	**Adenosine**, **digoxin**, atropine, **magnesium**, ivabradine	Adenosine: SVT • Mg: torsades

Amiodarone (most tested)

- **Class III** (but has **all 4 class** effects: Na, β, K, Ca blockade) • most effective antiarrhythmic, **safe in HFrEF and CAD** • **low torsades risk** despite QT ↑.
- **Dose:** oral loading **400 mg BID–TID** (800–1,600 mg/day) until **~10 g** total → maintenance **100–200 mg daily** • IV (stable VT/AF): **150 mg over 10 min → 1 mg/min × 6 h → 0.5 mg/min × 18 h** • cardiac arrest: 300 mg → 150 mg IV push.
- **Very long half-life (~40–60 days)**, highly lipophilic → effects and DDIs persist for months after stopping • contains **iodine**.
- **ADRs**: **pulmonary toxicity** (pneumonitis/fibrosis — most serious), **thyroid** (**hypothyroidism** > hyperthyroidism), **hepatotoxicity**, **corneal microdeposits** (common, benign), **optic neuropathy**, **blue-gray skin discoloration**, **photosensitivity**, bradycardia/AV block, QT ↑, peripheral neuropathy, tremor, ataxia • IV: **hypotension**, phlebitis (central line if > 2 mg/mL), bradycardia.

- **Monitoring: baseline** ECG, **TFTs**, **LFTs**, **CXR + PFTs** (with DLCO), eye exam if visual problems → **TFTs and LFTs every 6 months**, CXR yearly, ECG periodically, eye exam if symptoms.
- **DDIs (inhibits CYP2C9, 2D6, 3A4, P-gp): warfarin** (↓ warfarin dose **30–50%**, monitor INR) • **digoxin** (↓ digoxin dose **50%**) • **simvastatin max 20 mg**, lovastatin max 40 mg • ↑ DOACs (P-gp) • ↑ cyclosporine, tacrolimus • additive QT drugs • bradycardia with β-blockers, CCBs, **sofosbuvir** (severe bradycardia) • grapefruit.

Other class III agents

Drug	Key points
Dronedarone (Multaq)	Amiodarone analog **without iodine** (less thyroid/pulmonary toxicity) • 400 mg BID with meals • **boxed: CI in NYHA IV HF or recent decompensated HF** and **permanent AF** (↑ death, stroke, HF) • hepatotoxicity, ↑ SCr (inhibits creatinine secretion), QT, GI • **CYP3A4 substrate** (CI strong inhibitors/inducers) • ↑ digoxin (↓ dose 50%), dabigatran, statins (simvastatin ≤ 10 mg)
Sotalol	β-blocker + class III • **initiate in hospital with continuous ECG for ≥ 3 days (5–6 doses)** • dose by CrCl (**CI if CrCl < 40** for AF; ventricular dosing differs), QTc > 450 baseline CI • **torsades**, bradycardia, bronchospasm, HF • renal elimination
Dofetilide (Tikosyn)	Pure I(Kr) blocker • **REMS**: start **inpatient ≥ 3 days** with ECG/QTc monitoring • **dose by CrCl**: > 60 → **500 mcg BID**; 40–60 → 250 mcg BID; 20–40 → 125 mcg BID; **< 20 CI** • hold/reduce if QTc ↑ > 15% or > 500 ms • **CI with verapamil, HCTZ, cimetidine, trimethoprim, ketoconazole/itraconazole, prochlorperazine, megestrol, dolutegravir** • safe in HFrEF/CAD • torsades
Ibutilide (Corvert)	IV for **acute conversion** of AF/flutter (flutter > AF) • 1 mg over 10 min (repeat once) • **torsades** (~4%) → monitor ECG ≥ 4 h, pretreat with Mg

7. Specific Arrhythmias

Bradyarrhythmias

- **Sinus bradycardia**, **sick sinus syndrome**, **AV blocks** (1st degree — PR > 200 ms; 2nd degree Mobitz I (Wenckebach) / Mobitz II; 3rd degree — complete).
- Treat only if **symptomatic** (hypotension, syncope, dizziness, HF) → **atropine 1 mg IV q3–5 min (max 3 mg)** → transcutaneous pacing or **dopamine 5–20 mcg/kg/min** or **epinephrine 2–10 mcg/min** → **permanent pacemaker** (Mobitz II, 3rd degree, symptomatic sinus node dysfunction).
- **Drug-induced:** β-blockers, non-DHP CCBs, digoxin, amiodarone, dronedarone, sotalol, clonidine, ivabradine, **donepezil** (cholinesterase inhibitors), lithium, fingolimod, lacosamide → review and stop/reduce • overdose antidotes: β-blocker → glucagon; CCB → calcium, high-dose insulin; digoxin → DigiFab.

Supraventricular tachycardias (PSVT — AVNRT, AVRT)

- Sudden onset/termination palpitations • narrow-complex regular tachycardia (150–250 bpm).
- **Acute (stable): vagal maneuvers** (modified Valsalva, carotid sinus massage) → **adenosine 6 mg rapid IV push + 20 mL NS flush → 12 mg** (may repeat 12 mg) → IV **diltiazem/verapamil** or **β-blocker** • **unstable → synchronized cardioversion**.
- **Adenosine:** half-life < 10 s • ADRs: **transient asystole/AV block**, **flushing**, **chest pressure**, dyspnea, **bronchospasm** (caution asthma) • **↓ dose (3 mg)** with **dipyridamole**, **carbamazepine**, central line, **heart transplant** • **↑ dose/less effect** with **theophylline, caffeine**.
- **Long-term: catheter ablation** (first-line, curative) • or β-blocker, diltiazem/verapamil • flecainide/propafenone (no structural disease).
- **Wolff-Parkinson-White (WPW, pre-excitation):** delta wave, short PR • **AF with WPW (pre-excited, irregular wide QRS) → AVOID AV-nodal blockers** (**adenosine, β-blockers, CCBs, digoxin, IV amiodarone**) → ↑ conduction via accessory pathway → VF • use **procainamide** or **ibutilide**, or **cardioversion** • ablation definitive.

Ventricular arrhythmias

- **PVCs:** usually benign → treat if symptomatic (β-blocker, CCB) or high burden (ablation) • avoid class Ic post-MI (CAST).

- **Monomorphic VT with pulse, stable: amiodarone 150 mg IV over 10 min** (repeat) → infusion • or **procainamide** (20–50 mg/min; avoid with ↑ QT/HF) or **sotalol** • **unstable → synchronized cardioversion** • pulseless → defibrillation (ACLS).
- **Prevention of sudden cardiac death: ICD** (secondary prevention after VF/sustained VT; primary prevention LVEF ≤ 35%) • antiarrhythmics (amiodarone, sotalol) only as adjuncts to reduce shocks • β-blockers.

Torsades de pointes (polymorphic VT with long QT)

- **QTc > 500 ms** (or ↑ > 60 ms from baseline) = high risk • treatment: **magnesium sulfate 2 g IV** (even if Mg normal) • **stop QT-prolonging drugs** • correct **K (> 4)** and **Mg (> 2)** • **overdrive pacing** or **isoproterenol** (↑ HR shortens QT) • **unstable/pulseless → defibrillation**.
- **Risk factors: female sex**, **hypokalemia**, **hypomagnesemia**, hypocalcemia, **bradycardia**, HF/LVH, elderly, congenital long QT, high drug levels (renal/hepatic impairment, DDIs), **multiple QT drugs**, recent conversion from AF.
- **QT-prolonging drugs: antiarrhythmics** (class Ia, **III**: sotalol, dofetilide, ibutilide, dronedarone, amiodarone) • **antibiotics** (**macrolides**, **fluoroquinolones**, pentamidine, azoles, bedaquiline) • **antipsychotics** (**haloperidol IV**, **ziprasidone**, thioridazine, quetiapine) • **antidepressants** (**citalopram > 40 mg** [> 20 mg if > 60 years], escitalopram, TCAs) • **methadone** • **ondansetron**, droperidol, metoclopramide, domperidone • **hydroxychloroquine**, chloroquine • arsenic trioxide, some TKIs • donepezil, tacrolimus, loperamide (high dose), cocaine.

PART C — ATRIAL FIBRILLATION (AF)

8. AF Overview (ACC/AHA/ACCP/HRS 2023)

- Most common sustained arrhythmia • **irregularly irregular** rhythm, **no P waves**, variable ventricular rate • loss of atrial kick → ↓ CO • stasis in **left atrial appendage** → **thrombus** → **stroke** (5× risk).
- **Symptoms:** palpitations, fatigue, dyspnea, dizziness, ↓ exercise tolerance — or asymptomatic • **tachycardia-induced cardiomyopathy** if uncontrolled rate.
- **Risk factors:** age, **HTN**, HF, CAD, valvular disease, **obesity**, **OSA**, **alcohol**, diabetes, **hyperthyroidism**, CKD, smoking, athletes, surgery (post-op AF), pulmonary disease/PE, sepsis, **drugs** (theophylline, sympathomimetics, ibrutinib).
- **Classification (2023 stages):** Stage 1 at risk • Stage 2 pre-AF • **Stage 3 AF**: **paroxysmal** (terminates within 7 days), **persistent** (> 7 days, needs intervention), **long-standing persistent** (> 12 months), successful ablation • **Stage 4 permanent** (no further rhythm control attempts).
- **Work-up:** ECG, echo, **TSH**, BMP, CBC, LFTs • look for reversible causes.
- **Management pillars:** (1) **stroke prevention** (anticoagulation), (2) **rate control**, (3) **rhythm control**, (4) **risk-factor modification** ("SOS": stroke, optimize risk factors, symptom control).

Risk-factor modification (reduces AF burden)

- **Weight loss ≥ 10%** (overweight/obese) • **alcohol reduction/abstinence** • **physical activity** • treat **OSA** (CPAP) • BP control • glycemic control • smoking cessation • treat hyperthyroidism.

9. Stroke Prevention in AF

CHA2DS2-VASc	Points
C — Congestive heart failure (or LV dysfunction)	1
H — Hypertension	1
A2 — **Age ≥ 75**	**2**
D — Diabetes mellitus	1
S2 — **Prior stroke/TIA/thromboembolism**	**2**
V — Vascular disease (prior MI, PAD, aortic plaque)	1
A — Age 65–74	1

CHA2DS2-VASc	Points
Sc — Sex category (female)	1

- **Oral anticoagulation (OAC) recommended** if **CHA2DS2-VASc ≥ 2 in men or ≥ 3 in women** (annual stroke risk ≥ 2%) • **reasonable** if score **1 in men or 2 in women** (risk 1–2%) • **score 0 (men) / 1 (women) → no anticoagulation** • (ESC 2024 uses **CHA2DS2-VA** — sex removed: OAC if ≥ 2, consider if 1).
- **Decision applies to all AF types** (paroxysmal, persistent, permanent) and atrial flutter • based on stroke risk, **not on rhythm** (continue OAC after successful rhythm control/ablation if risk factors).
- **Aspirin (alone or with clopidogrel) is NOT recommended** for stroke prevention in AF (less effective, similar bleeding).
- **DOACs preferred over warfarin** (except **mechanical heart valves** and **moderate–severe mitral stenosis** → **warfarin**) • DOACs: ↓ ICH, similar/better efficacy.
- **Bleeding risk (HAS-BLED ≥ 3 = high)**: **H**TN (uncontrolled), **A**bnormal renal/liver function, **S**troke, **B**leeding history, **L**abile INR, **E**lderly (> 65), **D**rugs (antiplatelets/NSAIDs) or alcohol → used to **identify and modify bleeding risk factors**, **not to withhold OAC**.
- **Left atrial appendage occlusion (LAAO — Watchman)** for patients with **contraindication to long-term OAC** (bleeding).
- **Device-detected subclinical AF** (< 24 h): OAC may be considered in high risk.

DOAC	Standard AF dose	Dose reduction criteria	Notes
Apixaban (Eliquis)	**5 mg BID**	**2.5 mg BID if ≥ 2 of 3: age ≥ 80, weight ≤ 60 kg, SCr ≥ 1.5 mg/dL**	Used in ESRD/dialysis • least bleeding • strong 3A4 + P-gp inhibitors → ↓ 50% (avoid if already 2.5)
Rivaroxaban (Xarelto)	**20 mg daily with evening meal**	**15 mg daily if CrCl 15–50** • avoid CrCl < 15	Must take **with food** • once daily
Dabigatran (Pradaxa)	**150 mg BID**	**75 mg BID if CrCl 15–30** (US) • avoid CrCl < 15 • P-gp inhibitors + CrCl 30–50 → 75 mg BID (dronedarone, ketoconazole)	**Dyspepsia** • don't open capsules • original bottle • idarucizumab reversal • ↑ GI bleeding
Edoxaban (Savaysa)	**60 mg daily**	**30 mg if CrCl 15–50 • do NOT use if CrCl > 95** (↓ efficacy — boxed)	Once daily
Warfarin	Dose to **INR 2–3** (target 2.5)	—	**Mechanical valve** (INR 2.5–3.5 mitral), **mitral stenosis** • TTR > 65–70%

Anticoagulation around cardioversion

- **AF ≥ 48 h or unknown duration → therapeutic anticoagulation for ≥ 3 weeks before** cardioversion **OR TEE** to exclude LA thrombus (then cardiovert on anticoagulation) • **continue ≥ 4 weeks after** cardioversion in **all** patients (atrial stunning) → then long term according to CHA2DS2-VASc.
- **AF < 48 h:** low stroke risk → may cardiovert (anticoagulation around the procedure) • high risk (CHA2DS2-VASc ≥ 2 men/≥ 3 women) → anticoagulate before (ideally < 12 h safest) and continue long term.
- **Hemodynamically unstable → immediate synchronized cardioversion** (start anticoagulation as soon as possible).

Periprocedural management and special situations

- **DOACs: no bridging** — hold 1–2 days (low bleeding risk) to 2–4 days (high risk), longer for dabigatran with ↓ CrCl • **warfarin**: hold 5 days; **bridging with LMWH generally NOT recommended** (BRIDGE trial) except mechanical valves/very high thrombotic risk.
- **AF + ACS/PCI:** short triple therapy (≤ 1 week) → **DOAC + clopidogrel** up to 12 months → DOAC alone.
- **AF + stroke:** start OAC within days depending on stroke size (early initiation reasonable — ELAN) • ICH: individualized.
- **Pregnancy:** LMWH (or warfarin in 2nd trimester for mechanical valves per specialist) • avoid DOACs.

- **Atrial flutter**: same anticoagulation rules • rate control harder • **catheter ablation (cavotricuspid isthmus) first-line**.
- **Post-operative AF**: often transient; β-blockers; OAC per risk if persistent.
- **Reversal:** warfarin → vitamin K + 4F-PCC • dabigatran → idarucizumab • apixaban/rivaroxaban → andexanet alfa or 4F-PCC.

10. Rate Control vs Rhythm Control

Rate control

- **Target resting HR < 100–110 bpm** (**lenient < 110** acceptable if asymptomatic and LVEF preserved — RACE II) • stricter (< 80) if symptoms persist.
- **LVEF > 40%: β-blocker** or **non-DHP CCB** (diltiazem, verapamil) first-line • digoxin as add-on.
- **LVEF ≤ 40% (HFrEF): β-blocker** (carvedilol, metoprolol succinate, bisoprolol) and/or **digoxin** • **avoid non-DHP CCBs** • **amiodarone** if refractory/hypotension.
- **Acute (IV) rate control: diltiazem 0.25 mg/kg IV over 2 min** (repeat 0.35 mg/kg) → infusion 5–15 mg/h • **metoprolol 2.5–5 mg IV** over 2 min (up to 3 doses) • esmolol • **digoxin IV** or **amiodarone IV** in HF/hypotension • avoid diltiazem/verapamil in decompensated HF.
- **Digoxin**: controls resting HR but not during exercise → add-on or sedentary/HF patients • **AV node ablation + pacemaker** for refractory rate.
- **Pre-excited AF (WPW)**: **no AV-nodal blockers** → procainamide/ibutilide/cardioversion.

Rhythm control

- **Indications:** symptoms despite rate control, **HFrEF** (ablation ↓ mortality — CASTLE-AF), younger patients, first episode, difficult rate control, patient preference • **early rhythm control** (within 1 year of diagnosis) ↓ CV outcomes (EAST-AFNET 4).
- **Catheter ablation** (pulmonary vein isolation) = **first-line** option in selected patients (symptomatic paroxysmal AF, HFrEF) — more effective than drugs.
- **Pharmacologic cardioversion: flecainide**, **propafenone** (no structural disease), **ibutilide**, **dofetilide**, **amiodarone** (slower; structural disease) • **"pill-in-the-pocket"**: **flecainide 200–300 mg** or **propafenone 450–600 mg** once at onset (after safety tested in hospital) + β-blocker/CCB beforehand.
- **Electrical cardioversion**: synchronized DC shock (sedation).

Patient characteristics	Antiarrhythmic drug options to maintain sinus rhythm
No structural heart disease (normal LV, no CAD)	**Flecainide**, **propafenone**, **dronedarone**, **sotalol**, dofetilide (amiodarone last line)
Coronary artery disease (normal LVEF)	**Dronedarone**, **sotalol**, **dofetilide** (amiodarone) • **NO class Ic**
HFrEF	**Amiodarone**, **dofetilide** (catheter ablation preferred) • **NO** class Ic, dronedarone (NYHA III–IV/decompensated), sotalol (relative)
Significant LVH / HCM	Amiodarone, dofetilide, disopyramide (HCM), dronedarone
CKD	Amiodarone, dronedarone (adjust/avoid sotalol, dofetilide by CrCl)

- **Class Ic agents (flecainide, propafenone)** must be given with an **AV-nodal blocker** (β-blocker, CCB) → can convert AF to atrial flutter with **1:1 conduction** (very fast ventricular rate).
- **Amiodarone** = most effective but most toxic → reserve when others fail or in HFrEF • **dronedarone** less effective, **CI in permanent AF and NYHA IV/recent decompensated HF**.

11. Arrhythmia and AF Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Amiodarone	PO load 400 mg BID–TID to ~10 g → **100–200 mg daily** • IV 150 mg/10 min → 1 → 0.5 mg/min	**Pulmonary fibrosis**, **thyroid**, **liver**, corneal deposits, optic neuropathy, **blue-gray skin**, photosensitivity, bradycardia	**↓ warfarin 30–50%**, **↓ digoxin 50%**, simvastatin ≤ 20 • TFT/LFT q6 mo, CXR yearly • half-life ~2 months
Dronedarone	400 mg BID with meals	Hepatotoxicity, ↑ SCr, QT, GI	**CI NYHA IV/recent decompensated HF, permanent AF** • CYP3A4 • ↓ digoxin 50%
Sotalol	80 mg BID → 160 mg BID (CrCl-based)	**Torsades**, bradycardia, bronchospasm	**Inpatient start 3 days** • CI CrCl < 40 (AF), QTc > 450
Dofetilide	**500 mcg BID** (CrCl > 60) → 250 / 125 mcg BID	**Torsades**	**REMS, inpatient 3 days** • CI CrCl < 20 • **CI verapamil, HCTZ, cimetidine, trimethoprim, ketoconazole, dolutegravir**
Flecainide	50–150 mg BID • pill-in-pocket 200–300 mg	Dizziness, blurred vision, proarrhythmia	**CI structural heart disease/CAD/HFrEF** • + AV-nodal blocker
Propafenone	150–300 mg TID (ER 225–425 BID) • pill-in-pocket 450–600 mg	**Metallic taste**, dizziness, bronchospasm	Same CI as flecainide • CYP2D6 • ↑ digoxin, warfarin
Ibutilide	1 mg IV over 10 min (repeat once)	**Torsades**	Acute conversion • monitor 4 h • Mg
Procainamide	20–50 mg/min IV (max 17 mg/kg) → 1–4 mg/min	Hypotension, QT, **lupus**, agranulocytosis	WPW-AF, stable VT • NAPA (renal)
Lidocaine	1–1.5 mg/kg → 1–4 mg/min	CNS toxicity, seizures	VT/VF
Adenosine	**6 mg rapid IV → 12 mg**	Asystole (brief), flushing, chest pressure, bronchospasm	↓ dose with dipyridamole/carbamazepine/transplant • ↑ with theophylline/caffeine • not in pre-excited AF
Diltiazem	IV 0.25 mg/kg over 2 min → 5–15 mg/h • PO 120–360 mg	Hypotension, bradycardia, edema	**Avoid HFrEF** • CYP3A4 inhibitor (statins, DOACs)
Verapamil	IV 2.5–5 mg • PO 120–480 mg	**Constipation**, bradycardia	Avoid HFrEF • ↑ digoxin • CI with dofetilide
Metoprolol	IV 2.5–5 mg q5 min × 3 • PO tartrate 25–100 BID / succinate	Bradycardia, hypotension	Rate control (any LVEF)
Esmolol	500 mcg/kg → 50–300 mcg/kg/min	Hypotension	Ultra-short (esterases)
Digoxin	IV load 0.25 mg q2h (max 1.5 mg) or 0.125–0.25 mg PO daily	Toxicity (GI, visual, arrhythmias)	HFrEF rate control • ↓ dose with amiodarone
Magnesium sulfate	**2 g IV** (torsades)	Hypotension, flushing	Torsades
Atropine	**1 mg IV q3–5 min (max 3 mg)**	Anticholinergic	Symptomatic bradycardia
Apixaban	**5 mg BID** (2.5 if 2 of: ≥ 80 y, ≤ 60 kg, SCr ≥ 1.5)	Bleeding	OK in dialysis • 3A4/P-gp
Rivaroxaban	**20 mg with evening meal** (15 mg CrCl 15–50)	Bleeding	With food

Drug	Dose	Key ADRs	Key notes / interactions
Dabigatran	**150 mg BID** (75 mg CrCl 15–30)	**Dyspepsia**, GI bleed	Original bottle, don't open • idarucizumab
Edoxaban	**60 mg daily** (30 mg CrCl 15–50)	Bleeding	**Not if CrCl > 95**
Warfarin	INR **2–3** (mechanical mitral 2.5–3.5)	Bleeding, skin necrosis	Mechanical valves, mitral stenosis • many DDIs

Exam Traps and Frequently Tested Points

1. **HFrEF ≤ 40%**, HFmrEF 41–49%, **HFpEF ≥ 50%**, HFimpEF (continue GDMT) • **stages A–D never go backward**; NYHA I–IV can change.
2. **Sacubitril/valsartan ↑ BNP** → monitor with NT-proBNP • BNP ↑ in CKD, ↓ in obesity.
3. **Four pillars for HFrEF: ARNI (or ACEi/ARB) + β-blocker + MRA + SGLT2i** — start early, titrate to target • loop diuretics only for symptoms.
4. **ARNI:** 36-h washout from ACEi • CI angioedema, pregnancy • start 49/51 BID (24/26 if naive/eGFR < 30) → 97/103 BID.
5. **Only carvedilol, metoprolol succinate, bisoprolol** in HFrEF • start low go slow when euvolemic • carvedilol with food, target 25 BID (50 BID > 85 kg).
6. **MRA:** K < 5.0, eGFR > 30 • check K at 3 days, 1 week • spironolactone → gynecomastia (switch eplerenone) • eplerenone CYP3A4.
7. **SGLT2i (dapagliflozin/empagliflozin 10 mg)** for HFrEF, HFmrEF, HFpEF regardless of diabetes • eGFR ≥ 20–25.
8. **H-ISDN** for Black patients NYHA III–IV on GDMT or ACEi/ARB/ARNI intolerance • hydralazine lupus.
9. **Ivabradine:** sinus rhythm, HR ≥ 70 on max β-blocker, LVEF ≤ 35% • phosphenes, bradycardia • **vericiguat:** worsening HF, CI nitrates/PDE-5i.
10. **Digoxin:** ↓ hospitalizations, not mortality • level 0.5–0.9 • toxicity: N/V, yellow-green vision, arrhythmias • ↑ by hypoK/hypoMg • **amiodarone → ↓ digoxin 50%** • DigiFab.
11. **IV iron** if ferritin < 100 or 100–299 + TSAT < 20% • **ICD** LVEF ≤ 35% • **CRT** LVEF ≤ 35% + LBBB QRS ≥ 150.
12. **Avoid in HFrEF:** NSAIDs, diltiazem/verapamil, TZDs, saxagliptin/alogliptin, class Ic, dronedarone (NYHA IV/recent decompensation), itraconazole, cilostazol.
13. **Loop equivalence: furosemide 40 PO = 20 IV = torsemide 20 = bumetanide 1** • furosemide ~50% bioavailable • metolazone for resistance • ethacrynic acid for sulfa allergy.
14. **ADHF:** warm/wet → IV diuretics (1–2.5× oral dose IV) ± vasodilators • cold/wet → inotropes (dobutamine, milrinone — renal adjust, hypotension) • continue β-blocker unless shock.
15. **Nitroprusside → cyanide/thiocyanate toxicity** • HF education: daily weights (> 2–3 lb/day or 5 lb/week), sodium restriction, avoid NSAIDs.
16. **HFpEF:** SGLT2i, diuretics, BP control, MRA/finerenone, ARNI/ARB (may consider), GLP-1/tirzepatide in obesity.
17. **Class Ic (flecainide, propafenone) CI in structural heart disease/CAD (CAST)** • give with AV-nodal blocker • pill-in-pocket.
18. **Amiodarone:** pulmonary, thyroid, liver, eyes, skin (blue-gray), photosensitivity • TFT/LFT every 6 months • ↓ warfarin 30–50%, ↓ digoxin 50%, simvastatin ≤ 20.
19. **Dofetilide and sotalol:** start inpatient 3 days, CrCl dosing, torsades • dofetilide CI with verapamil, HCTZ, cimetidine, trimethoprim, ketoconazole, dolutegravir.
20. **Dronedarone:** CI permanent AF and NYHA IV/recent decompensated HF • procainamide → lupus, NAPA • disopyramide anticholinergic.
21. **Adenosine 6 → 12 mg rapid push** • ↓ dose with dipyridamole, carbamazepine, transplant, central line • theophylline/caffeine ↓ effect • bronchospasm.
22. **WPW with AF: avoid AV-nodal blockers** (adenosine, BB, CCB, digoxin, IV amiodarone) → procainamide/ibutilide/cardioversion.
23. **Torsades → Mg 2 g IV**, stop QT drugs, K > 4, Mg > 2, overdrive pacing/isoproterenol • QTc > 500 = high risk • know QT drug list.

24. **Symptomatic bradycardia:** atropine 1 mg (max 3 mg) → pacing/dopamine/epinephrine • drug causes: BB, CCB, digoxin, amiodarone, clonidine, donepezil.
25. **CHA2DS2-VASc:** age ≥ 75 and stroke = 2 points • **OAC if ≥ 2 men / ≥ 3 women** • consider 1 men / 2 women • **aspirin not recommended**.
26. **DOACs preferred** except mechanical valves and moderate–severe mitral stenosis (warfarin) • **HAS-BLED** modifies risk factors, doesn't exclude OAC • LAAO if OAC contraindicated.
27. **DOAC AF doses:** apixaban 5 BID (2.5 if 2 of 3: ≥ 80, ≤ 60 kg, SCr ≥ 1.5) • rivaroxaban 20 with evening meal (15 if CrCl 15–50) • dabigatran 150 BID (75 if CrCl 15–30) • edoxaban 60 (30 if CrCl 15–50; not if > 95).
28. **Cardioversion:** AF ≥ 48 h → 3 weeks OAC before (or TEE) + **4 weeks after for everyone** • unstable → immediate cardioversion.
29. **Rate control target < 100–110** • LVEF > 40%: BB or diltiazem/verapamil • **LVEF ≤ 40%: BB ± digoxin (no non-DHP CCB)** • IV diltiazem 0.25 mg/kg.
30. **Rhythm control:** ablation first-line in selected (HFrEF, paroxysmal) • no structural disease → flecainide/propafenone/dronedarone/sotalol • CAD → dronedarone/sotalol/dofetilide • HFrEF → amiodarone/dofetilide.
31. **No bridging for DOACs** • warfarin bridging only for mechanical valves/high risk • AF + PCI → DOAC + clopidogrel.
32. **AF risk-factor modification:** weight loss ≥ 10%, alcohol reduction, OSA treatment, exercise, BP control • check TSH.

Note: doses and recommendations follow the 2022 AHA/ACC/HFSA HF guideline with 2024 ECDP updates, ESC 2023, and the 2023 ACC/AHA/ACCP/HRS AF guideline. If your course slides or reference use slightly different numbers, **use your course numbers for the exam**.